

Tehman Hassan

Final Year Computer Engineering Student
Department of Computer Engineering, University of Engineering
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2023-UET/CP-011
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Profile and Bio

Final-year Computer Engineering student at UET Taxila specializing in backend systems, distributed architecture, and AI-powered application development. Experienced in designing and deploying scalable REST APIs, real-time WebSocket systems, and microservices-based architectures using Python, Django, FastAPI, and AWS — validated through a software engineering internship, freelance client deployments, and multiple hackathon-winning projects.

Education

Bachelor of Engineering in Computer Engineering

Aug 2023 - Present

University of Engineering and Technology Taxila, Pakistan

Currently in 6th Semester, having an updated CGPA of 3.38

Secondary & Higher Secondary Education

Apr 2017- Oct 2022

Cadet College Choa Saida Shah Chakwal, Pakistan

A+ Grade · Computing Club · Football Team · Debating Society · Appointment Holder

Projects

Personalized LMS System – 1st Runner Up in AI Hackathon, May 2026

Project Link: <https://atomcamp.duckdns.org/>

- Built an adaptive learning platform that personalizes student paths in real time based on performance data
- Implemented dynamic quiz difficulty adjustment and instructor analytics dashboard
- Stack: Python, FastAPI, React, OpenAI API, PostgreSQL

Built “talentgraph.ai”, talent mapping in Global AI Hackathon, 2026

Project Link: <https://skillpath.duckdns.org/>

- Built a global skill-mapping platform that clusters talent by niche, location, and market trends
- Integrated AI-based job description parsing to match organizations with relevant talent pools
- Stack: Python, Django, React, OpenAI API, PostgreSQL

Full Stack Health Platform for Luther Health LTD (Mar 2025 – Sep 2025)

Project Link: <https://luther.health/>

- Developed a production-grade health management SaaS for a UK-based doctor with 15+ assessment funnels
- Reduced AI infrastructure costs by architecting an n8n workflow automation layer on AWS
- Stack: Python, Django, AWS EC2, n8n, OpenAI GPT-4, React, Stripe, PostgreSQL

Single Cycle MIPS Pipelined Processor Datapath in Verilog HDL (Jan 2026 – Feb 2026)

- Designed a full single-cycle pipelined datapath supporting R, I, Load, Store, and Branch instruction types
- Stack: Verilog HDL, Vivado

Real-Time Asynchronous Chat App Using WebSockets with End-to-End Encryption (Jul 2025 – Sep 2025)

Project Link: <https://github.com/Tehman700/Real-Time-Chat-App-2-Way-Communication-WebSockets>

- Built a full-stack async chat application with E2E encryption, OTP verification, and offline message handling
- Stack: Python, Django Channels, WebSockets, PostgreSQL, React

Potholes Complaint System (Winning Project) in OOP Course Exhibition (Oct 2024 – Dec 2024)

- Built an Android app using Azure Vision AI to verify pothole photo authenticity before filing complaints
- Designed a Spring Boot backend with PostgreSQL for complaint tracking and admin resolution management
- Stack: Java, Spring Boot, Azure Vision AI, PostgreSQL

AI Agent for Teachers based on Blooms Taxonomy Level (Jan 2024 – Apr 2024)

Project Link: https://github.com/Tehman700/Dr.Fawad_AI_Agent_Course

- Built a Java-based AI agent that analyzes academic documents and auto-generates CLO-aligned reports
- Stack: Java, OpenAI GPT-4, PDF generation

Work Experience

Freelance Backend Software Developer

Aug 2025 – Jan 2026

Fiverr, Freelance Platform

- Built and deployed production-grade REST APIs using Python, Django REST Framework, and PostgreSQL, successfully delivering projects for 5+ clients across multiple time zones
- Implemented real-time features using Django Channels and WebSockets, enabling bi-directional communication across 3+ client applications
- Designed and maintained CI/CD pipelines on AWS EC2 with Nginx reverse proxy, cutting manual deployment effort by ~40% and improving release reliability
- Achieved Level 1 Seller status within 2 months, maintaining a 5-star rating across all delivered orders through consistent delivery and systematic debugging
- Integrated subscription billing flows and automated activity logging using Django signals, reducing manual tracking overhead across client platforms

Software Engineering Backend Intern

Jul 2025 – Sep 2025

Inara Technologies, F11 Markaz, Islamabad, Pakistan

- Architected a full-stack REST API from scratch using Django REST Framework, implementing JWT authentication, role-based access control, and custom exception handling across 10+ modules
- Built a nested commenting system supporting likes, dislikes, and threaded replies with full CRUD operations across 5+ content modules
- Developed real-time two-way communication using WebSockets and Django Channels, enabling live user interaction between clients and server
- Configured GitHub Actions CI/CD pipelines integrated with AWS, automating build and release workflows and reducing manual deployment steps by ~50%

Skills

- **Languages:** Python, Java, JavaScript, Verilog, SQL
- **Frameworks:** Django REST, React, FastAPI, Node.js, Django Channels
- **Cloud & DevOps:** AWS EC2, S3, Nginx, GitHub Actions, Docker
- **Tools:** n8n, Postman, Git, WebSockets
- **Databases:** PostgreSQL, MongoDB

Competitions

1st Runner Up in AI Hackathon at Aurex'26 at Bahria H11

Placed 1st Runner Up at Aurex'26 AI Hackathon, building an AI-powered adaptive LMS with team Titans, powered by atomcamp

Winner OOP Course Semester Project

Won the best Semester Project in my department among 150 students, along with certificates and prize money

Participation in Hackathon in Devsinc

This was a Speed Programming Competition in a Professional Software Company, tackling Data Structures, Greedy Algorithms, and DP Problems

Certificates

- Software Development with GitHub Copilot
- API Testing Learning Path Certificate
- Postman API Fundamentals Student Expert
- Machine Learning with Python
- 100 Days of Code: The Complete Python Pro Bootcamp
- SQL Basic Hacker rank
- Transformers in LLMs
- Thinking and Problem Solving (LinkedIn)